

L4

Intermediate loops

Join Discord - <https://tinyurl.com/ly-discord>

Conditional statement $\rightarrow$ Decision

Repeat $\rightarrow$ Loops

For $\rightarrow$ start to end

While $\rightarrow$ digits

sum of digits

print digits

1 ... n

% $\rightarrow$ Remainder

Even $\rightarrow$ $\% 2 == 0$

(Print all numbers

$\% 6 == 0$ or

$\% 11 == 0$)



Divisible by 11

for (i = 1 ; i <= N ; i++)

{

if (i % 6 == 0 || i % 11 == 20)

{

S.O. Plu (i) ;

}

}

loop (For)

Practice Question [Recap]

Print "FizzBuzz" if i is divisible by 3 and 5. X

Print "Fizz" if i is divisible by 3.

Print "Buzz" if i is divisible by 5.

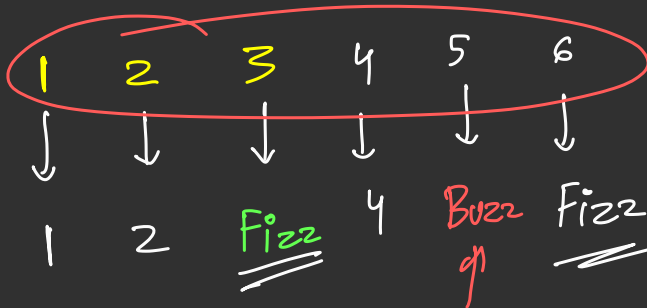
Print i if none of the above conditions are true.

$$\underline{\underline{3 \% 3 = 0}}$$

↑
1 - - - n

15
↓

Fizz Buzz



if (i % 3 == 0 & & i % 5 == 0)

{
 s.o.pln ("Fizz Buzz");

}

else if (i % 3 == 0)

{
 s.o.pln ("Fizz");

}

else if (i % 5 == 0)

{
 s.o.pln ("Buzz");

}

else

{
 s.o.pln (i);

}

i → for



1 ... n

Loop stop

Break and continue

Jump to next iteration

Let's say in fizzbuzz, an extra condition is that for multiples of 4, do nothing. What can be done in this scenario?

Also if there is a danger number also given, how can we stop the loop at that time?

17

n = 25

danger number

17 → stop

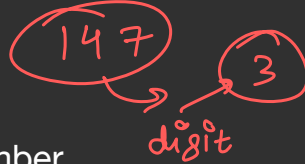
stop

break

while loop

Loops Continued

Count the number of digits of a positive number.



Find last digit $\rightarrow \% 10$ } $\rightarrow$ Sat Recording
Remove last digit $\rightarrow / 10$

$$n = 123$$

$$n = \frac{123}{10} = 12$$

$$n = \frac{12}{10} = 1$$

while ($N > 0$)
{

$$n = n / 10 ;$$

$$\text{count} = \text{count} + 1 ;$$

}

int count = ~~0~~;

~~3~~

$\frac{1}{10} = \underline{\underline{0}}$

while ($N > 0$) stop

$n = n / 10;$

count = count + 1;

3

3

S.O. Plu (count);

→ 4

3 → 1
↓
4

$N = \cancel{4156} = 0$

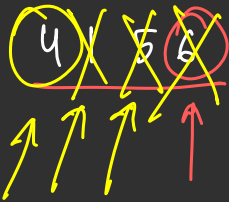
↑ ↑

Ans = 4

④

count = 0 ✓

$N = \frac{4156}{10} = \frac{415}{10} = \frac{41}{10}$
 $= \frac{4}{10} = 0$



Loops Continued

$$N = 4156$$

$$\downarrow 16$$

A sum of digits of a positive number.

```
while ( n > 0 )
```

```
{
```

```
    int lastDigit = n % 10;
```

```
    sum = sum + lastDigit;
```

```
    n = n / 10;
```

```
}
```

$$\text{sum} = 0 + 6 = 6$$

$$\text{sum} = 6 + 5 = 11$$

$$\text{sum} = 11 + 1 = 12$$

$$\text{sum} = 12 + 4 = \underline{\underline{16}}$$

S.O.Pln (sum);

Loops Continued

Sum of digits of all numbers from 1 to N.

for loop

1 - - - - n

while loop

1 → sum = 1

2 → sum = 2

9 → sum = 9

10 → sum = 1

11 → sum = 2

for (initialization; test condition; update)

initialization

do
{

statement }

update expression;

} while (test condition);

Hello
Hello
Hello



int i = 1;

3 <= 3

do
{

→ s.o.println ("Hello");

→ i++;

while (i <= 3);



False

~~i = 4~~

$i = i + 1$

$\text{int } i = 2;$

Even
number

infinite

$\{$
for (; ;)

S.O. Pln (i);

$i = i + 2;$

if (i > n)
break;

$\}$

increment by

2

break

↓

stop

$n = 15$

Loops Continued

For without initialization / condition / updation

Example of printing even numbers from 1 to N using a for loop, but without doing initialization / condition / updation in the loop.

While Loop Question

→ 5 ~~k~~ = 6

Get First N numbers that have sum of digits = k

6	15	24	33	42
↓	↓	↓	↓	↓
6	6	6	6	6

While Loop Question

Check whether a number is palindrome or not

Practice and Homework Questions

- Check if a given number is a power of 2 or not
 - <https://leetcode.com/problems/power-of-two/>
- Check if a given number is prime or not?
- Print all prime numbers from 1 to N.
- Calculate product of all digits
- Print all factors of a number
- Print first and last digit of a number

Thank You!

Please practice more questions and examples as above !!